Ocean Route Optimization for the Vendée Globe via Reinforcement Learning

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1 Abstract

This work presents an autonomous route planning system for the Vendée Globe race, the renowned single-handed, non-stop sailing circumnavigation of the globe. The adopted approach is based on Reinforcement Learning: an agent trained with the Proximal Policy Optimization (PPO) algorithm learns to control a simulated IMOCA 60 class sailing boat within a custom Gymnasium environment, fed by real meteorological data from the ECMWF ERA5 reanalysis dataset.

The agent receives a 30-feature observation vector that encodes the vessel's state, current weather conditions, wind forecasts at three temporal horizons, and geometric navigation indicators. Training was carried out over two historical racing seasons (2016–2017 and 2020–2021) for approximately 15 million steps, and evaluation was conducted on a season never seen during training (2024–2025), in order to measure the agent's generalization capability.

Results show that the agent is able to complete the circumnavigation with a high finish rate on the test season, autonomously navigating around the three great capes (Cape of Good Hope, Cape Leeuwin, and Cape Horn) without the need for manually hardcoded geographic penalties.

2 Introduction

Optimal route planning in a maritime environment is a sequential control problem characterized by high meteorological uncertainty, complex physical constraints, and a temporal horizon spanning several months. In the context of ocean races such as the Vendée Globe, the problem acquires an additional layer of difficulty: the boat must circumnavigate the globe single-handed, making the best use of continuously evolving wind conditions while adhering to strict geographic rules, without any external support.

Classical weather routing techniques rely on heuristic search algorithms applied to meteorological forecast grids. Such approaches, while effective in structured scenarios, prove fragile when faced with unforeseen conditions and require highly detailed physical models of the vessel. Reinforcement Learning (RL), by contrast, enables the learning of a control policy directly through interaction with the environment, adapting to variable conditions without explicit programming of the desired behaviour.

In this project we build a custom simulation environment that faithfully reproduces the dynamics of the Vendée Globe:

- real ERA5 wind data and a 6-hour temporal resolution, trilinearly interpolated during simulation;
- a physical boat model based on IMOCA 60 polar diagrams;
- seven mandatory geographic waypoints corresponding to the official race route.

3 Background

3.1 The Vendée Globe

The Vendée Globe is a single-handed, non-stop, unassisted sailing race, considered one of the most extreme sporting endeavours in the world. The start and finish line is located at Les Sables-d'Olonne, on the Atlantic coast of France. Competitors must circumnavigate the globe from west to east, passing mandatorily south of the three great capes: Cape of Good Hope (southern Africa), Cape Leeuwin (south-western Australia), and Cape Horn (Chile). The first edition was held in 1989 and the race takes place every four years. The course record stands at approximately 64 days, set in the 2024 edition. Environmental conditions are extreme: participants face storms in the Southern Ocean, Antarctic ice, and complete isolation for more than two months.

3.2 IMOCA 60 boats

The boats used in the Vendée Globe belong to the IMOCA 60 class (60 feet in length, $\approx 18.3\,\mathrm{m}$). These are high-performance ocean monohulls, characterized by:

- carbon composite hulls weighing less than 8 tonnes;
- canting keels that reduce heeling and increase upwind speed;
- lateral foils that allow the boat to fly above the water surface, reaching speeds exceeding 30 knots.

In our project, the boat's behaviour is modelled using an IMOCA 60 polar diagram interpolated with bicubic splines, scaled by a realism factor of 0.85 to account for average sailing conditions.

3.3 Terminology

Understanding the system requires familiarity with concepts from two distinct fields: sailing navigation and Reinforcement Learning applied to maritime routing. The following tables collect the most relevant terms used throughout the rest of this document.

3.3.1 Sailing terminology

Term	Definition
TWS (*True Wind Speed*)	Absolute wind speed relative to the sea, expressed in knots. Independent of the boat's motion.
TWD(*True Wind Direction*)	The direction from which the absolute wind originates, expressed in degrees relative to geographic north.
TWA (*True Wind Angle*)	The angle between the boat's bow and the absolute wind direction. It determines the point of sail: close-hauled, reaching, or running.
No-go zone	An angular sector centered on the wind direction (typically ± 40–$45°$) in which a sailboat cannot move forward because the sails cannot generate lift. Entering it causes speed to drop to 0–3 knots.
Tacking	A turning maneuver where the bow crosses the wind direction to change tack while sailing upwind. It allows progress against the wind using a zigzag trajectory.
Gybing	A maneuver similar to tacking, but with the stern crossing the wind direction. Used during downwind points of sail.
Polar diagram	A graph describing the maximum theoretical speed of a boat as a function of TWA and TWS. It represents the vessel's aerodynamic and hydrodynamic performance.
VMG (*Velocity Made Good*)	The component of the boat's speed projected in the direction of the target waypoint. It measures the geographic efficiency of the route: a high VMG indicates real progress toward the destination, regardless of absolute speed.

4 Related Work

While dynamic weather routing has been studied extensively using classical deterministic algorithms, the specific application of Deep Reinforcement Learning (DRL) to global, long-horizon regattas like the Vendée Globe remains a nascent field. The most direct and relevant predecessor to our proposed methodology is the work of Mannocci (2023) [2], who pioneered the use of DRL specifically for navigating the Vendée Globe course.

In his formulation, Mannocci addressed the meteorological routing problem by translating localized satellite wind data into synthetic image channels. These spatial representations of the wind field surrounding the vessel were processed using a Convolutional Neural Network (CNN) to extract environmental features. The routing task itself was framed as a discrete Markov Decision Process; the agent's control mechanism was restricted to a highly quantized action space consisting of only three possible outputs: turn clockwise, turn anticlockwise, or maintain the current heading.

Although Mannocci successfully demonstrated the feasibility of training

an RL agent for global circumnavigation, a critical analysis reveals several structural limitations inherent to that architecture. Primarily, forcing a vessel's trajectory into a discrete action space inherently introduces severe quantization errors. In physical maritime navigation, discrete steering commands result in jagged, inefficient trajectories that bleed forward momentum and fail to smoothly exploit the highly non-linear aerodynamic gradients of an IMOCA 60 polar profile. Furthermore, utilizing CNNs to process gridded spatial images of wind introduces significant computational overhead and relies on spatial discretization, which poorly approximates the continuous fluid dynamics of ocean weather. Finally, reacting primarily to a static spatial snapshot of the current wind field limits the agent's temporal foresight, which is a critical requirement for anticipating large-scale frontal passages over a multi-month race.

Our proposed architecture builds directly upon this conceptual foundation but resolves these specific limitations by transitioning the entire simulation pipeline into a strictly continuous, differentiable domain. Rather than employing a discrete action space, our formulation utilizes the Proximal Policy Optimization (PPO) algorithm to output a fully continuous rudder adjustment, bounded mathematically to replicate realistic physical turn-rate limits.

To provide environmental awareness, we replace the computationally heavy CNN spatial extraction with rigorous trilinear interpolation (across latitude, longitude, and time) of the ERA5 dataset. This yields a dense, mathematically continuous 30-feature state vector that natively includes the continuous bicubic spline evaluation of the vessel's polar diagram. Crucially, our state space augments immediate weather awareness with rolling, multi-horizon temporal forecasts ($t + 6$, $t + 12$, and $t + 24$ hours). This paradigm shift allows the policy to proactively anticipate temporal meteorological shifts and execute smooth, continuous long-term strategies, representing a significant architectural advancement over discrete, spatially-reactive routing models.

5 Methodology

This section details the formulation of the autonomous sailing problem as a continuous Markov Decision Process (MDP) and describes the simulation environment, data integration, and reinforcement learning architecture employed to solve it.

5.1 Meteorological data integration

To provide the reinforcement learning agent with a physically accurate and highly dynamic meteorological context, the simulation environment integrates historical atmospheric data from the ERA5 reanalysis dataset, provided by the European Centre for Medium-Range Weather Forecasts (ECMWF). ERA5 was selected as the foundational weather model due to its high spatiotemporal resolution and its status as the gold standard for historical climate data, making it uniquely suited for modeling the complex, global-scale weather systems encountered during the

Vendée Globe circumnavigation. Specifically, the environment ingests the u_{10} and v_{10} atmospheric variables, which represent the eastward and northward wind velocity vectors at ten meters above the ocean surface. These components are critical, as they allow the environment to mathematically derive both the True Wind Speed (TWS) and True Wind Direction (TWD) at any given moment, establishing the primary environmental forcing that dictates the vessel's kinematic potential.

Incorporating global, high-resolution meteorological data spanning multiple multi-month racing seasons introduces a massive memory footprint, which can severely bottleneck the rapid environment sampling required for reinforcement learning. To resolve this computational constraint and ensure efficient data throughput during training, the raw GRIB atmospheric arrays are systematically preprocessed and cast to a half-precision floating-point format (float16 instead of float32). This dimensionality optimization dramatically reduces the RAM requirements and accelerates state-space array queries. The inherent minor loss in extreme numerical precision is negligible within the macroscopic context of global weather routing, yet the computational tradeoff is strictly necessary to allow the agent to experience millions of simulated timesteps within a feasible training duration.

Furthermore, a fundamental misalignment exists between the discrete, gridded nature of historical weather datasets and the continuous spatiotemporal state space in which the autonomous vessel operates. If the agent were to query wind data solely from the nearest discrete grid points or hourly timestamps, it would experience artificial, instantaneous jumps in wind velocity (the dataset has new data every six hours). Such discontinuities severely disrupt the Markov Decision Process, leading to erratic reward signals, unstable gradient updates, and ultimately, poor policy convergence. To bridge this gap and simulate a continuous, fluid environment, a rigorous trilinear interpolation mechanism is implemented. By interpolating across three dimensions—latitude, longitude, and time—the environment dynamically computes precise, localized wind vectors for the exact coordinates and timestamp of the vessel. This spatiotemporal smoothing ensures that the agent experiences realistic wind gradients, gradual frontal passages, and continuous meteorological transitions, thereby enabling the policy to learn proactive, long-horizon routing strategies rather than merely reacting to abrupt, artificial data shifts

5.2 Environment dynamics and physical modeling

The simulation requires a high-fidelity physics and geospatial engine to accurately translate the agent's routing decisions and the ambient atmospheric conditions into realistic maritime kinematics over a global scale. This section outlines the mathematical models governing the vessel's movement and its spatial relationship to the Earth.

5.3 Geospatial coordinate modeling

The environment fundamentally operates on a spherical coordinate system to faithfully represent the Earth's geometry, a prerequisite for accurately simulating a circumnavigation task. Euclidean distance metrics are inherently distorted and inadequate for global-scale navigation. Therefore, all spatial calculations are strictly computed using the Haversine formula. This ensures that the agent interacts with accurate great-circle distances, forcing the policy to learn true global routing optimizations rather than exploiting planar projection distortions.

Furthermore, a critical component of the simulation's validity is its robust collision detection and boundary enforcement mechanism. The environment integrates a global land mask to define terrestrial boundaries, requiring the agent to actively anticipate and navigate around continents, capes, and islands. In addition to terrestrial obstacles, the simulation mathematically enforces the Antarctic Ice Exclusion Zone, a crucial safety constraint in the real-world Vendée Globe race. This is implemented as a strict navigational floor at a latitude of -60.0 degrees. Breaching either the land mask or the ice boundary acts as a failure state, resulting in an immediate terminal penalty and the cessation of the episode. This dual-constraint system forces the agent to develop precise spatial awareness and risk-averse maneuvering alongside pure speed optimization.

5.4 Vessel kinematics and polar modeling

The physical response of the autonomous agent to the wind is parameterized by a performance model specific to the IMOCA 60 racing class utilized in the Vendée Globe. To model this accurately, the vessel's theoretical velocity is dictated by its polar performance diagram [1], a standardized maritime matrix that maps a given True Wind Speed (TWS) and True Wind Angle (TWA) to an expected boat speed. Within the simulation loop, the TWA is dynamically calculated at each timestep as the absolute angular difference between the vessel's continuous heading and the interpolated True Wind Direction (TWD).

However, raw polar diagrams provide only a discrete grid of speed values. In a continuous-state reinforcement learning environment, relying on discrete look-up tables or nearest-neighbor approximations introduces artificial step-functions into the agent's velocity. These discontinuities destabilize the calculation of the reward function and misrepresent the fluid dynamics of sailing, preventing the policy from learning fine-grained steering adjustments. To resolve this, the discrete IMOCA 60 polar matrix is mapped into a continuous, differentiable mathematical surface using a 2-dimensional bicubic spline interpolation. This continuous formulation allows the simulation engine to output highly precise and smooth boat speeds for any arbitrary, fractional combination of wind angle and velocity.

5.5 Reinforcement Learning formulation

The autonomous navigation task is rigorously formulated as a continuous-space Markov Decision Process (MDP), characterized by its state representation, action bounds, and a dense, multi-objective reward signal. However, framing a 24,000-nautical-mile circumnavigation as a single, monolithic reinforcement learning objective introduces extreme sparse-reward challenges, making policy convergence nearly impossible. To render the global route computationally tractable and provide dense, directional feedback to the agent, the macroscopic race is mathematically decomposed into a sequential waypoint navigation task. The environment defines a deterministic sequence of crucial geographical gates corresponding to the real-world Vendée Globe trajectory (e.g., passing the Cape of Good Hope, Cape Leeuwin, and Cape Horn). At any given timestep, the agent's MDP is exclusively conditioned on a single active waypoint. The simulation continuously monitors the distance between the vessel and this active target. Once the vessel successfully breaches a predefined acceptance radius around the target coordinates, the environment dynamically shifts the active objective to the subsequent waypoint in the sequence. This sequential targeting mechanism ensures that the agent is perpetually presented with a localized, solvable optimization problem, which sequentially chains together to accomplish the global circumnavigation.

5.5.1 Observation space

The observation space is designed to provide the agent with complete situational awareness, encompassing a 30 dimensional continuous feature vector. A fundamental prerequisite for the stable optimization of deep reinforcement learning policies—particularly when utilizing Multi-Layer Perceptrons (MLPs)—is the rigorous normalization and encoding of the input observation space. The raw simulated state variables exist across wildly disparate scales: distances span thousands of nautical miles, wind speeds range from 0 to over 50 knots, and geographic coordinates are bound between ± 180 degrees. If fed directly into the neural network, these unscaled values would disproportionately bias the gradient descent updates, leading to unstable convergence, vanishing gradients, or policy collapse. Consequently, all continuous scalar features (such as distance, velocity, and positional coordinates) are systematically normalized to a strict [-1, 1] interval based on their theoretical maximum bounds.

Furthermore, cyclical variables, most notably the vessel's heading, True Wind Direction (TWD), and True Wind Angle (TWA), present a unique mathematical challenge. In a raw scalar format, angular representations suffer from severe numerical discontinuities (e.g., the transition from 359 degrees to 0 degrees represents a minimal physical rotation but a massive numerical leap). To resolve this, all angular features are mapped into a continuous 2-dimensional Cartesian space using trigonometric encoding. Each angle θ is decomposed into its continuous sine and cosine components ($\sin\theta$, $\cos\theta$). This transformation guarantees that the neural network interprets the angular state space as a smooth,

continuous manifold, which is vital for learning fine-grained steering policies.

Finally, the state vector is logically partitioned into four distinct semantic domains:

1. **Vessel kinematics:** This domain captures the vessel's instantaneous physical status and its precise global context. It includes the absolute latitude and longitude, which ground the agent within the Earth's spherical coordinate system. Furthermore, it incorporates the vessel's current speed over ground and its trigonometric heading, defining its immediate momentum and motion vector. Crucially, the current rudder angle is also ingested, providing the network with awareness of its internal mechanical state. Finally, the active waypoint is included, acting as the localized optimization target that gives immediate directional purpose to the rest of the kinematic variables.

2. **Current wind data:** This domain encapsulates the immediate atmospheric forcing acting upon the vessel at the current simulation timestep. It includes the raw u10 and v10 wind components, TWA, TWD and TWS.

3. **Forecasts:** Every 3 hours of simulation timestep, the environment queries the ERA5 atmospheric model for future atmospheric states at intervals of $t+6$, $t+12$, and $t+24$ hours. Specifically, the environment projects the agent's current geographical coordinates forward in the dataset's time dimension, extracting the anticipated u_{10} and v_{10} wind vectors and computing the TWS, TWD and difference between the current TWS and the future one. By ingesting these temporally shifted wind data, the PPO agent learns to execute preemptive maneuvers, mirroring the complex meteorological routing utilized by human navigators.

4. **Land and ice distances:** Spatial awareness of critical hazards is essential for the survival of the agent. The state space explicitly includes quantifiable proximity metrics to both terrestrial obstacles and the Antarctic ice exclusion zone, allowing the neural network to learn safety margins dynamically before a terminal collision occurs. The proximity to the ice exclusion zone is computationally straightforward due to its definition as a static latitudinal boundary. It is computed as the one-dimensional latitudinal delta between the vessel's current position and the strict -60.0 degree floor.

   Conversely, computing the exact continuous distance to complex global coastlines is highly computationally intensive. To maintain the rapid environment sampling efficiency required for reinforcement learning, the distance to the nearest landmass is not calculated at every timestep; rather, it is estimated and updated only once every five simulation steps. In order to compute that, the environment projects a 4x4 grid of coordinate probe points spanning a 1.5-degree radius around the vessel's current latitude and longitude. The simulation then evaluates each of these sixteen test coordinates against a high-resolution binary global land mask. If a

probe point intersects with a registered landmass, the great-circle distance from the vessel to that specific coordinate is calculated. The minimum recorded distance across all intersecting probe points is then extracted as the coastal proximity metric, capped at a maximum observable threshold of 500 nautical miles. By incorporating these proximity gradients directly into the observation space, the policy network is provided with dense feedback regarding its spatial safety margins.

5. **Efficiency metrics:** Includes the vessel's real-time Velocity Made Good on Course (VMC) toward the active waypoint. In sailing kinematics, a vessel's absolute speed through the water is often misleading; a boat sailing at 25 knots in the wrong direction is making negative progress. VMC isolates the specific vector component of the boat's velocity that is directly closing the distance to the target destination. By explicitly providing the VMC to the neural network, the agent receives immediate, dense feedback regarding the geographic efficiency of its current heading. This allows the policy to rapidly distinguish between high-speed reaching angles that actively advance the race and high-speed angles that merely parallel or diverge from the optimal great-circle route.

5.5.2 Action space

The agent operates within a 1-dimensional continuous action space bounded by a strict [-1.0, 1.0] interval. In the context of maritime navigation, formulating the control mechanism as a continuous space is vastly superior to discrete action branching (e.g., hard-coded commands to turn port, turn starboard, or hold course). Discrete control introduces severe quantization errors, forcing the vessel into jagged, inefficient trajectories that bleed momentum and fail to exploit the continuous, differentiable aerodynamic gradients of the IMOCA 60 polar performance surface. Crucially, the policy's output does not dictate an absolute global heading. Instead, the continuous scalar is mathematically mapped to a relative rudder adjustment, capping the maximum heading alteration at ± 20 degrees per one-hour simulation timestep. This relative formulation is structurally and physically significant. If the agent were permitted to output absolute headings directly, the model could theoretically execute instantaneous, physically impossible 180-degree turns within a single timestep. Furthermore, predicting absolute cyclic angles introduces severe boundary discontinuities during neural network optimization, destabilizing the gradient descent process. Consequently, this restricted action space mathematically forbids hyper-reactive, instantaneous evasion. Because the agent cannot simply spin the boat instantly to avoid a hazard or catch a sudden gust of wind, it is forced to leverage the multi-horizon wind forecasts and spatial proximity metrics provided in the observation space. The bounded action space essentially mandates proactive strategic planning, requiring the policy to initiate gradual, calculated maneuvers hours in advance of weather fronts and coastlines, thereby successfully replicating the long-horizon decision-making of expert human navigators.

5.5.3 Reward function

In complex spatial navigation tasks, the design of the reward function is the most critical factor in aligning the agent's optimization landscape with the actual physical objectives of the simulation. For the autonomous sailing problem, a monolithic, sparse reward granted only upon race completion is entirely insufficient for policy convergence due to the extreme length of the temporal horizon. Therefore, the reward signal is heavily shaped as a dense, multi-objective function.

The primary driving force of the policy is the progress reward, which is directly proportional to the net geographical distance closed toward the active waypoint at each timestep. By mathematically linking the primary positive reward to the Velocity Made Good (VMG) rather than absolute vessel speed, the MDP prevents the agent from exploiting high-wind scenarios that result in rapid motion but geographic divergence. The agent is forced to learn the delicate aerodynamic trade-off between sailing at a sub-optimal heading to gain massive wind-speed advantages (reaching) versus sailing directly toward the waypoint at a slower velocity.

To refine the agent's behavioral policy and ensure mechanical viability, several auxiliary dense signals are incorporated:

- **Time and efficiency penalty:** A constant, negative baseline penalty is applied at every simulation timestep. This universal temporal friction naturally incentivizes the agent to complete the route as expediently as possible, implicitly discouraging cyclical trajectories or loitering in favorable but unproductive wind zones.

- **Kinematic stability penalty:** Continuous control algorithms are notoriously prone to high-frequency action jitter. In a physical maritime context, continuous, violent rudder oscillations induce massive hydrodynamic drag and stress the vessel's actuators. To enforce a mechanically viable policy, a stability penalty subtracts a scalar value proportional to the magnitude of the delta between the current and previous rudder actions, forcing the network to learn smooth, deliberate steering transitions.

Finally, the reward function must rigidly enforce the strict safety constraints of the Vendée Globe through a hierarchy of spatial penalties:

- **Navigational constraint:** The environment applies dynamic, moderate penalties when the vessel enters perilous and narrow but technically navigable regions, such as the turbulent coastal waters of Cape Horn. This encourages the agent to establish a proactive safety margin before an absolute collision risk arises.

- **Land and ice constraints:** The most severe penalties are reserved for absolute failure states: crashing on the coast or crossing the -60.0 degree Antarctic ice exclusion zone. Triggering either of these conditions instantly halts the simulation and applies a massive, discrete negative reward (e.g.,

-500). This catastrophic penalty ensures a steep negative gradient at the boundaries. When coupled with the spatial proximity metrics provided in the observation space, the agent rapidly learns to associate the narrowing proximity gradients with this terminal consequence, ultimately prioritizing vessel survival over marginal speed optimizations.

5.6 Agent architecture and training

To map the high-dimensional, continuous state space to the bounded continuous action space, the Proximal Policy Optimization (PPO) algorithm was selected. In the context of global maritime routing, the environment is characterized by highly non-linear dynamics—specifically the IMOCA 60 polar splines and the shifting gradients of the ERA5 weather systems. Traditional Deep Q-Networks (DQN) are incapable of handling the continuous control required for nuanced rudder adjustments, while standard policy gradient methods (such as REINFORCE or standard Actor-Critic) often suffer from catastrophic performance collapse if a gradient update steps too far into a sub-optimal policy space. PPO mitigates this by utilizing a clipped surrogate objective function, ensuring that policy updates are strictly bounded. This trust-region optimization is paramount in our environment, where a single poor policy update could cause the agent to consistently crash into land or the ice zone, irrevocably poisoning the value function with massive terminal penalties.

5.6.1 Network architecture

Both the actor (policy) and critic (value) networks were parameterized using Multi-Layer Perceptrons (MLPs). Given the explicit, dense nature of the engineered observation space (30 fully observable, normalized features), complex spatial extractors like Convolutional Neural Networks (CNNs) or temporal models like Long Short-Term Memory (LSTMs) were deemed unnecessary and computationally prohibitive. Instead, a shared architecture comprising two hidden layers of 256 units each was deployed. This [256, 256] configuration provides sufficient representational capacity to learn the complex trigonometric mappings and multi-horizon wind correlations without inducing severe overfitting to the training weather patterns.

5.6.2 Training strategy and environmental randomization

A fundamental challenge in dynamic weather routing is preventing the agent from simply memorizing a static, historically optimal route. If trained on a single meteorological season, the neural network would overfit to the specific timing of low-pressure systems and frontal passages of that specific year, resulting in a brittle policy incapable of zero-shot generalization.

To enforce the learning of robust meteorological heuristics rather than route memorization, the training regimen employed temporal domain randomization. The agent was trained over a total of 15,000,000 environment timesteps. For each

training episode, the environment's initialization logic randomly sampled starting parameters from two entirely distinct historical Vendée Globe racing seasons: the 2016-2017 season and the 2020-2021 season. By continuously shuffling the underlying ERA5 temporal index, the agent was forced to experience a highly diverse distribution of wind data.

5.6.3 Monitoring and callbacks

Due to the length of the training horizon and the severity of the terminal collision penalties, the training process was augmented with custom programmatic callbacks. A continuous evaluation callback monitored the agent's rolling episodic rewards. To prevent the degradation of a high-performing policy—a common issue in continuous RL when an agent temporarily explores a highly penalized sub-space—the best-performing weights were systematically checkpointed. Furthermore, an early-stopping mechanism was introduced; if the mean rolling reward fell beneath a critical collapse threshold (e.g., -700), indicating that the policy had degenerated into cyclical crashing behaviors, the training loop was immediately truncated to conserve computational resources.

6 Experimental results

Evaluation was carried out over 100 simulation episodes on the 2024–25 test season, completely unseen during training, using real ERA5 wind data starting from the official race start date (10 November 2024).

6.1 Outcome distribution

The 100 episodes fall into three distinct categories:

- **FINISH** (80 episodes, 80%): the boat completes the circumnavigation passing through all seven mandatory waypoints and the three great capes, with an average time of approximately **85 days**. All three gates (Cape of Good Hope, Cape Leeuwin, Cape Horn) are correctly crossed.
- **Early Do Not Finish (DNF) — No-Go Zone** (11 episodes, 11%): the boat correctly leaves Les Sables-d'Olonne but, after a few hundred steps in the North Atlantic, enters a headwind zone (*no-go zone*) from which it cannot escape. Speed drops to 2–3 knots and the episode terminates upon exhaustion of the maximum step count without having reached WP1.
- **Late Do Not Finish (DNF) — No-Go Zone** (9 episodes, 9%): the boat completes the full circumnavigation (WP1–WP6, all gates), but during the final northward leg of the Atlantic toward Les Sables-d'Olonne the agent again enters a no-go zone off the Iberian Peninsula without reaching WP7.

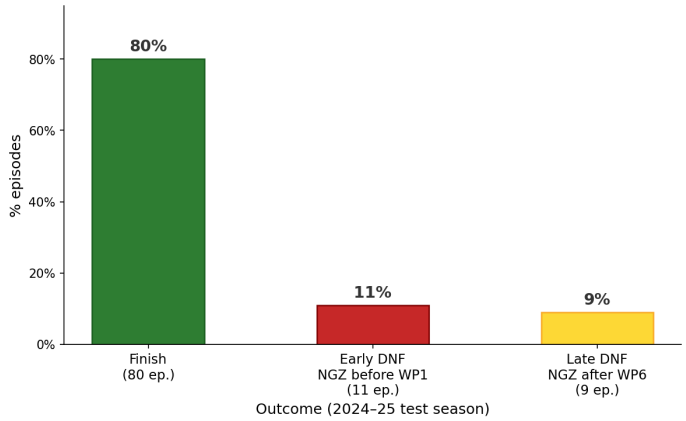

Figure 1: Distribution of the 100 episodes on the 2024–25 test season.

6.2 Error analysis

Early DNF. The failure before WP1 is attributable to insufficient learning during training. The agent did not develop a sufficiently robust *tacking* strategy for prolonged close-hauled conditions, and consequently remains trapped in the no-go zone until the episode ends. This category represents a case of failure to generalize to out-of-distribution wind patterns.

Late DNF. The failure after WP6 is qualitatively different: the agent demonstrates the ability to circumnavigate the globe successfully, but fails to complete the final approach. In these episodes, the bearing toward WP7 requires a north-north-east climb through the Bay of Biscay with often easterly or north-easterly winds. The agent, which during training predominantly encountered the northward return leg with favourable Atlantic westerlies, appears to misidentify the geometry of the final stretch and deviates to the west. This behaviour is consistent with insufficient exposure to adverse wind configurations during the multi-season return leg training.

Common root cause: action temporal granularity. Both failure modes share a structural limitation of the simulation environment: the interaction timestep is fixed at $\Delta t = 1\,\text{hour}$, meaning the agent can adjust its heading at most once per simulated hour. In reality, a sailor can perform tacks and gybes as frequently as necessary, adapting in real time to gusts and wind direction changes. With $\Delta t = 1\,\text{h}$, the agent has too coarse a temporal resolution to react promptly to short-lived adverse wind configurations, remaining trapped in situations from which a human navigator would escape with rapid, successive manoeuvres.

A possible solution is to reduce the timestep to $\Delta t = 30\,\text{min}$ or $\Delta t = 10\,\text{min}$, increasing the available control frequency. This direction is discussed in Section 7,

Conclusions and Future Work.

6.3 Comparison between training and test results

Figure 2 summarises the outcome distribution across 100 episodes per simulation. During the simulation with training winds, each episode randomly selected one of the two seasons (2016–17 or 2020–21). This yielded 52 episodes on the 2020–21 season and 48 episodes on the 2016–17 season.

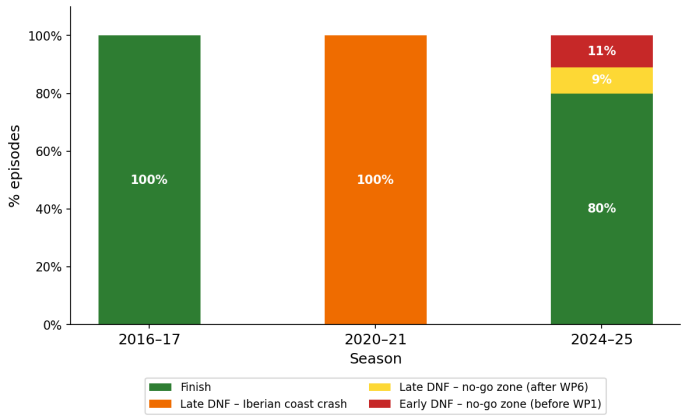

Figure 2: Percentage distribution of outcomes by season.

The most relevant result is the behaviour on the 2020–21 season, one of the training seasons: the agent achieves a 0% finish rate, with all 52 episodes ending in a crash against the Iberian coast around simulated day 79.

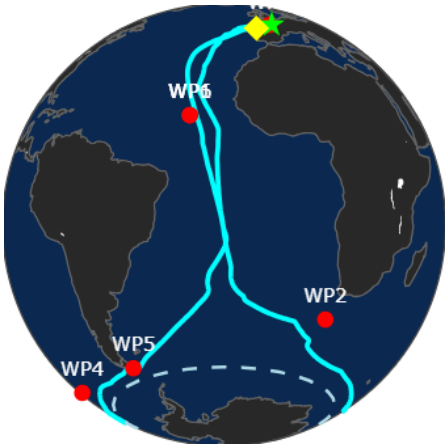

Figure 3: DNF episode 2020–21: full trajectory ending in a crash on the Iberian coast.

Figure 4: DNF episode 2020–21: close-up of the final leg showing the deviation toward the coast.

This is not a generalised failure: in all episodes the agent correctly completes the full circumnavigation, passing the three great capes and reaching WP6. The problem manifests exclusively in the final leg, the northward Atlantic return to Les Sables-d'Olonne, where the 2020–21 winds appear to exhibit a particularly adverse north-easterly configuration that pushes the agent toward the coast instead of allowing the climb north.

Paradoxically, the model generalises better on the unseen 2024–25 test season (80% finish rate) than on one of its own training seasons. This rules out the overfitting hypothesis for the 2020–21 season and suggests that the agent has learned a sufficiently general policy, but that the 2020–21 season presents anomalous Atlantic return conditions relative to the learned distribution. The 2016–17 season confirms this reading, with a 100% finish rate indicating that the policy is fully adequate for standard wind conditions on the final leg.

6.4 Interactive visualisation

All episodes from the test phase (2024–25 season) described above are viewable on the project's interactive web page, available at:

`https://alessandrocampedelli.github.io/VendeeGlobeRL/`

The page displays the 3D orthographic globe with real simulation trajectories (derived from the model's GPS data), episode cards with per-episode statistics, and a summary panel showing the outcome distribution.

7 Conclusions and future work

This work has demonstrated that a Reinforcement Learning agent trained with the PPO algorithm on real ERA5 meteorological data is capable of autonomously learning a navigation policy for the Vendée Globe circumnavigation, without any explicit geographic supervision and without manually hardcoded penalties for the critical sections of the route.

The final model, trained on two historical seasons (2016–17 and 2020–21) for approximately 15 million steps, achieves an 80% completion rate on the unseen 2024–25 test season. The agent autonomously learns to navigate around the three great capes, to avoid the Antarctic exclusion zone, and to exploit favourable winds across the different phases of the route.

Analysis of the failure modes revealed that incomplete episodes fall into two distinct categories — an early DNF due to inability to escape the no-go zone, and a late DNF due to deviation on the final Atlantic leg — both sharing a structural limitation of the interaction timestep ($\Delta t = 1\,\mathrm{h}$), which reduces the agent's ability to react promptly to adverse wind conditions. The comparison with the training seasons further highlighted that the 2020–21 season presents anomalous Atlantic return conditions relative to the learned distribution, ruling out overfitting and confirming the generalization robustness of the policy.

7.1 Future work

1. Reduction of the integration timestep. The common root cause of the main observed failure modes is the temporal granularity of the action, fixed at $\Delta t = 1\,\mathrm{hour}$. Reducing the step to $\Delta t = 30\,\mathrm{min}$ or $\Delta t = 10\,\mathrm{min}$ would allow the agent to react more promptly to adverse wind configurations, bringing the simulated behaviour closer to that of a real navigator who can tack as frequently as needed. This direction was explored during project development but was abandoned due to the proportional increase in episode length and the number of training steps required for convergence, resulting in computational times incompatible with the available hardware resources.

2. Integration of ocean currents and wave physics. The current model considers wind as the sole environmental variable, neglecting ocean currents and wave physics. In reality, currents such as the Gulf Stream, the Antarctic Circumpolar Current, and equatorial currents significantly influence both the speed and the optimal route of an IMOCA 60. Integrating ocean current datasets (e.g., CMEMS — Copernicus Marine Service) and a wave resistance model into the vessel speed equation would make the simulation environment more physically accurate, potentially improving both the quality of the learned policy and the transferability of the results to real operational scenarios.

3. Reduction of mandatory waypoints. The current route is structured around seven explicit geographic waypoints that guide the agent along the official race track. This design choice, motivated by the need to simplify the

problem during early development, limits the generality of the learned policy: the agent learns to sequentially reach predefined targets rather than autonomously planning a global route. A promising direction is the progressive reduction of the number of mandatory waypoints, up to the extreme case where the only explicit objective is to return to the starting point after circumnavigating the globe while respecting the three mandatory gates. This would require the agent to develop a deeper understanding of route geometry and global meteorological patterns, producing a more generic and potentially more robust policy for unseen wind conditions.

4. Extension to a multi-agent competitive environment. The current architecture trains a single agent navigating the environment in isolation, strictly optimizing its trajectory against the simulated meteorological conditions. However, the Vendée Globe is fundamentally a competitive regatta where a skipper's long-term strategic decisions are continuously influenced by the positions, speeds, and projected routes of rival vessels. Transitioning the problem from a single-agent formulation to a Multi-Agent Reinforcement Learning (MARL) framework would allow for the simulation of complex fleet dynamics. By introducing the relative coordinates and states of competing autonomous vessels into the observation space, the agents would be challenged to learn advanced racing tactics—such as covering opponents, defending leads, or taking calculated meteorological risks to overcome a deficit. This extension would elevate the model from a pure weather routing optimizer to a comprehensive competitive sailing simulation, bringing it significantly closer to the true game-theoretic nature of the race.

References

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